DERIVED QUASI-COHERENT OBJECTS ARE THEIR DERIVED LIMITS OF TRUNCATIONS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Higher cohomology of quasi-coherent modules on affine opens vanishes. invoked at proof lines 365-369. Verifies the hypothesis of the limit-dimension criterion.
- The limit-dimension criterion identifies an object of the derived category with its derived limit of canonical truncations when the cohomology sheaves have vanishing higher cohomology on the affine cover. invoked at proof lines 370-371. Concludes the isomorphism.

Source reference. Original remark 0H72: sources/stacks-chow/tags.txt, source lines 21442–21442. Cited at statement lines 355-357 as the map whose isomorphism is claimed; the proof is inside the counted span.

Source reference. Original lemma 070M: sources/stacks-chow/tags.txt, source lines 8961–8961. Cited in the footnote at proof lines 359-360 for the K-injective consequence; the isomorphism argument is inside the counted span.

Source reference. Original lemma 01XB: sources/stacks-chow/tags.txt, source lines 2512–2512. Cited at proof lines 368-369 to verify the vanishing hypothesis; the conclusion is inside the counted span.

Source reference. Original lemma 0D64: sources/stacks-chow/tags.txt, source lines 16509–16509. Cited at proof lines 370-371 to conclude; the vanishing verification is inside the counted span.

Lemma 0.2. *Let X be a scheme. Let E be an object of $D_{QCoh}(\mathcal{O}_X)$. Then the map $E \to R\lim \tau_{\geq -n}E$ of Derived Categories, Remark 0H72 is an isomorphism*[1]*.*

Proof. Denote $\mathcal{H}^i = H^i(E)$ the ith cohomology sheaf of E. Let $\mathcal{B}$ be the set of affine open subsets of X. Then $H^p(U, \mathcal{H}^i) = 0$ for all $p > 0$, all $i \in \mathbf{Z}$, and all $U \in \mathcal{B}$, see Cohomology of Schemes, Lemma 01XB. Thus the lemma follows from Cohomology, Lemma 0D64. □

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[1] In particular, E has a K-injective representative by Derived Categories, Lemma 070M.